
Heena Khan

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gitHub: heenakhan122 **Portfolio:** heenakhan122.github.io/Heena-Khan-Portfolio/

EDUCATION

Stanford University

Expected Graduation: June 2027(B.S)

B.S. Computer Science (Artificial Intelligence) | Minor: Biology

Relevant Coursework: Operating Systems, Compilers, Data Structures & Algorithms, Databases, Probability & Statistics, Machine Learning, Discrete Mathematics, Cloud Computing, Computer Vision

EXPERIENCE

Software Engineering Intern – Data Infrastructure & Full Stack

Roche/Genentech

Santa Clara, CA | June 2026 – October 2026

- Architected PostgreSQL database with multi-column indexing to store and query PCR and NGS assay records across Roche's diagnostic portfolio, reducing query latency by [X]%
- Built domain-specific AI agent over live assay data, enabling natural language querying of [X] PCR records across [Y] assay types by internal scientists
- Engineered pre-ingestion drift detection pipeline monitoring schema validity and value distributions across ETL workflows, preventing silent data corruption in downstream agent responses
- Developed full-stack Django + React application with real-time PCR amplification curve visualizations, deployed via Docker with automated CI/CD validation

AI Research Engineering Intern

Neurotrack Technologies

Palo Alto, CA | June 2025-September 2025

- Engineered high-throughput data curation and feature pipelines in Python/SQL, ingesting and normalizing 10,000+ multimodal assessment records across 9 heterogeneous input schemas.
- Designed automated validation and noise-filtering algorithms to detect schema drift and artifact corruption, increasing input dataset reliability for downstream training pipelines.
- Trained and evaluated linear regression and parametric curve-fitting models in Python to map score distribution trajectories, using statistical evaluation metrics to prune 6 redundant assessment variants.

Stanford Tech Fellowship

Pull for Progress

Stanford, CA | June 2023 - September 2024

- Engineered geospatial machine learning models in Python using multidimensional health metrics to predict regional risk profiles across 50,000+ healthcare data points.
- Applied statistical modeling and spatial feature extraction to optimize healthcare worker deployment algorithms in resource-limited settings.
- Built reproducible data processing pipelines using Pandas and NumPy to clean, validate, and analyze high-dimensional public health datasets for strategic decision-making.

PROJECTS

AMANI – AI-Enabled Health Mapping System | Stanford Healthcare Design Challenge Finalist

React Native, TensorFlow Lite, PostgreSQL, REST APIs, Computer Vision

- Led development of a mobile application integrating on device computer vision to analyze visual health related inputs in low connectivity environments.
- Built user facing workflows translating model outputs into intuitive visual feedback within a React Native interface.
- Optimized mobile inference pipeline under latency and memory constraints to preserve smooth interaction.
- Designed backend APIs and structured database models supporting application functionality.

ModestFilter AI – Chrome Extension (Manifest V3)

JavaScript, Chrome APIs, Web Workers, Computer Vision, ML

- Designed and implemented browser extension integrating real time image processing into an interactive UI.
- Built modular frontend architecture supporting reusable components and iterative feature expansion.
- Implemented asynchronous logic while preserving responsive and intuitive user experience.
- Conducted live user testing sessions and refined UI interactions based on feedback.

SKILLS

Languages: Python, Java, C++, SQL, JavaScript/TypeScript

Systems & Tools: Linux, Git, Bash, PostgreSQL, MySQL

Concepts: Concurrency, Multithreading, Memory Management, Compiler Design, Relational Data Modeling, REST APIs